Kanghe Lyu

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Date of Birth: November 2003

Education

2022.09- Present **Sichuan University** ***School of Mathematics*** **Mathematics (Honors Program))**

Core Courses & Grades: Analytic Geometry (91)，Mathematical Modeling and Experimentation (96)，Number Theory and Algebraic Foundations (90)，Real Analysis (94)，Abstract Algebra (90)，Functional Analysis (93)

GPA: 3.71/4.0 **Major Ranking: 11/40** **English Proficiency: CET-4 (609), CET-6 (568)**

Academic exchange

2024 New Uzbekistan University Summer Program: Integrable Systems Related to Reflection Groups in Algebra, Geometry, and Topology

Key Takeaways: I met many like-minded peers and gained insights into the motivations and fundamental concepts of algebraic topology and differential manifolds. This experience deepened my awareness of both the vastness of mathematics and the gaps in my own expertise, motivating me to focus more intently on deep learning and reflection in mathematics, continuously improving my mathematical thinking abilities.

Research & Competition

2023 China Undergraduate Mathematical Contest in Modeling: "Optimal Heliostat Field Design Using Nonlinear Programming" National Second Prize
Introduction: Abstracted practical problems into geometric objects, established a heliostat field coordinate system, and performed standardized testing. Built a geometric optical model based on Euclidean theory to determine optimal solutions through heliostat distribution simulation.
Contributions: As team leader, coordinated tasks, abstracted geometric objects, simplified models to reduce computational complexity, and guided paper writing.
2024 National Innovation and Entrepreneurship Training Program for College Students
"Lei Zhenzi - Bridging Anomaly Detection and Traceability System for Multi-Source Heterogeneous Data in Smart Grids"
Introduction: Developed an anomaly detection system for smart grid data using biological immune bridging principles, featuring: 1) Heterogeneous detector generation; 2) Immune detector bridging; 3) Bridging-based detection/traceability models and prototype.
Contributions: As core member: 1) Designed implementation plan (immune modeling, clustering methods, detector splicing); 2) Prepared proposal (literature review, grid security analysis, model refinement).

Student Affairs

Sep 2022 - Sep 2024 Class Monitor, Class 6 of 2022 Cohort, School of Mathematics, Sichuan University

Sep 2023 - Sep 2024 Captain, Badminton Team, School of Mathematics, Sichuan University

Honors & Awards

National Level:

- 2023 China Undergraduate Mathematical Contest in Modeling (Team Leader) – *National Second Prize*
- 2024 National College Student Innovation and Entrepreneurship Training Program
  "Lei Zhenzi – Bridging Anomaly Detection and Traceability System for Multi-Source Heterogeneous Data in Smart Grids" – Excellent Completion

Provincial Level:

- 2023 14th China Undergraduate Mathematics Competition – *Provincial Second Prize*
- Sichuan Comprehensive Quality Assessment Certificate (Grade A)

University Level:

- China International College Student Innovation Competition (2024) – *Sichuan University Silver Award*
- Sichuan University 2023-2024 Academic Year – *Comprehensive First-Class Scholarship (Top 5%)*
- 2023-2024 Fall Semester "Little Spark" Research Program (Issue 28)
  "Properties and Applications of Gamma Function" (Project Leader) – *Completed*
- Sichuan University 2022-2023 Academic Year – *Outstanding Student Cadre*
- Sichuan University 2023-2024 Academic Year – *Outstanding Undergraduate Student*
- Sichuan University 2024 "Youth in Motion" Oriented Relay Race – *First Prize*